

# Memo 01: Mass-Conservation Review of `conc_a` and `conc_d`

Codex review for FVCOM-MP flocc diagnostics

April 23, 2026

## 1 Purpose

This memo summarizes the present numerical treatment of the flocc-enabled plastic tracers `conc_a` (aggregated microplastic) and `conc_d` (disaggregated microplastic) in `mod_plast.F`, with a focus on mass conservation when the interaction term is active. It also records the leading hypotheses for the observed mass loss.

## 2 Present Numerical Treatment

### 2.1 State Variables

With `PLAST_FLOC = T`, the water-column plastic state is split into two tracers:

$$C = C_a + C_d,$$

where  $C_a$  is stored in `conc_a` and  $C_d$  is stored in `conc_d`. The bed state is represented by the total areal mass `mass` together with split bookkeeping `mass_a` and `mass_d`.

### 2.2 Operator Sequence in the Current Code

The actual chronology in `Advance_plast` is summarized below.

| Checkpoint       | Current treatment                                                                                                                                                                                                                                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step start       | The code rebuilds the combined concentration <code>conc = conc_a + conc_d</code> , and then enters the bed-exchange operators.                                                                                                                                       |
| After deposition | Settling is solved separately for <code>conc_a</code> and <code>conc_d</code> . Aggregated plastic uses a flocc/sediment settling velocity, while disaggregated plastic uses the plastic settling velocity. The stored depositional flux is only the combined scalar |

$$\text{dflx} = \text{dflx}_a + \text{dflx}_d.$$

No separate bed fluxes are retained for the split states.

| Checkpoint                   | Current treatment                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| After erosion                | The bed erosion flux <code>eflx</code> is first computed as a total plastic flux. It is then partitioned back into <code>conc_a</code> and <code>conc_d</code> using the instantaneous bottom-layer ratio $r_b = \frac{\text{conc\_a}(k_{bm1})}{\text{conc}(k_{bm1})},$ with a fallback to purely disaggregated input if the denominator is too small. A similar split is used for the surface precipitation flux. Both additions are passed through local growth limiters. |
| After<br>Update Bottom       | The total bed mass is updated through the net bed exchange $\Delta M_b \propto \text{eflx} - \text{dflx},$ but the split bed masses are not updated from separate depositional fluxes. Instead, the current bed split ratio is reused to repartition the updated total mass. In other words, the bed split evolves by bookkeeping ratio, not by resolved split deposition.                                                                                                  |
| After<br>advection/diffusion | <code>conc_a</code> and <code>conc_d</code> are advected independently by two calls to <code>adv_scal</code> , then vertically diffused independently by two calls to <code>vdif_scal</code> .                                                                                                                                                                                                                                                                              |
| After kinetics               | The flocculation interaction step applies the pairwise update $C_a^{n+1} = C_a^* + \alpha C_d^* - \beta C_a^*$ and $C_d^{n+1} = C_d^* - \alpha C_d^* + \beta C_a^*$ , with $\alpha = \min(\lambda_a \Delta t, 1)$ and $\beta = \min(\lambda_d \Delta t, 1)$ . Algebraically, this step preserves $C_a + C_d$ pointwise before later clipping.                                                                                                                               |
| After<br>clamps/limiters     | Two non-conservative corrections remain in the flocculation path: an upper cap <code>cnew_a</code> > 100 or <code>cnew_d</code> > 100, and the final nonnegative clip <code>conc_a</code> = <code>max(cnew_a, 0)</code> and <code>conc_d</code> = <code>max(cnew_d, 0)</code> . The final combined concentration is then rebuilt from the clipped split fields.                                                                                                             |

### 2.3 Initialization and Restart Notes

- Initial water-column concentration is split by `INI_CONC_AG_RATE`.
- Initial bottom mass is now split consistently by `INI_MASS_AG_RATE` for both constant and static bottom-mass initialization paths.
- Hot-start support still restores only the combined plastic concentration in the current restart reader, so flocculation-state restart consistency remains a concern.

## 3 Conservation Expectations

If external sources and open-boundary forcing are disabled, the following operators should be internally conservative in total plastic mass:

- separate settling plus bed update, provided the water-column loss exactly matches the bed gain;

- separate erosion plus bed update, provided the bed loss exactly matches the water-column gain;
- the pairwise aggregation/disaggregation kinetics, because the gain in one split state is the loss in the other;
- interior advection and diffusion, up to the conservation properties of the transport scheme.

Therefore, any drift in the internal total

$$\text{water total} + \text{bed total}$$

should point to clipping, limiter truncation, split-state bookkeeping inconsistency, or an operator mismatch between the water-column and bed updates.

## 4 Leading Hypotheses for the Observed Mass Loss

### 1. Floc-only hard caps and clips are explicit sinks or gains

The upper cap on `cnew_a` and `cnew_d` removes positive mass directly. The nonnegative clip removes negative values, which acts as a positive mass correction.

### 2. Erosion and surface growth limiters can decouple bed and water updates

The erosion and surface additions are limiter-controlled in the water column, but the bed update still uses the full total fluxes `eflx` and `dflx`. If the limiter truncates the water-column increment while the bed still loses the full amount, the difference appears as an artificial sink.

### 3. Reuse of one `cflx` array across both tracer solves may distort floc-mode OBC handling

Both `conc_a` and `conc_d` currently call `adv_scal` with the same `cflx` storage. This may be harmless for strictly interior transport, but it is a plausible source of distorted boundary-flux bookkeeping when open boundaries are active.

### 4. Bed split bookkeeping is not driven by split deposition

The total bed mass is updated with the combined net bed flux, but `mass_a` and `mass_d` are then reconstructed from a ratio rather than from resolved split deposition and erosion fluxes. That makes the split bed inventory diagnostic rather than prognostically exact, and it can hide where the mismatch is generated.

## 5 Suggested Diagnostic Interpretation

A future diagnostic pass should report domain-integrated values for:

- water `a`, water `d`, water total;
- bed total, bed `a`, bed `d`;
- checkpoint-to-checkpoint totals at: `step_start`, `after_deposition`, `after_erosion`, `after_update_bottom`, `after_advdiff`, `after_kinetics`, and `after_clamps`;

- explicit correction counters for upper-cap loss, nonnegative-clip gain, erosion-limiter truncation, surface-limiter truncation, and bed-split reconciliation.

For an internal-isolation test with no point sources and no open-boundary forcing, the most useful checks are:

1. the total change from `after_advdiff` to `after_kinetics`, which should be near zero if the kinetic exchange is behaving conservatively;
2. the total change from `step_start` to `after_clamps`, which should be explained by the reported correction counters.

## 6 Bottom Line

The interaction step itself is algebraically conservative; the stronger candidates for significant mass loss are the flocc-only truncation operators and the mismatch between total bed fluxes and split-state water-column limiters/bookkeeping. These checkpoints are the first places to inspect in a later diagnostic pass.